MIL-S-13727B
1 February 1983
SUPERSEDING
MIL-S-13727A
30 April 1974

MILITARY SPECIFICATION

SODIUM PHOSPHATE, MONOBASIC, ANHYDROUS, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of anhydrous monobasic sodium phosphate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-S-48 - Sacks, Shipping, Paper
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)

MILITARY

MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Chemical characteristics. The anhydrous monobasic sodium phosphate shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. Chemical characteristics

Characteristic	Requirement	Test paragraph
Assay as monobasic sodium phosphate (NaH_2PO_4), minimum percent by weight	98.0	4.2.4.1
Insoluble matter, maximum percent by weight	0.1	4.2.4.2
pH of a 1.0-percent solution	4.6 ± 0.1	4.2.4.3

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the anhydrous monobasic sodium phosphate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 100 grams (g) shall be removed from each sample container and placed in a suitable clean, dry, moisture-proof container labeled to identify the lot and container from which it was taken.

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TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container, ready for shipment. Sample shipping containers shall be examined for the following defects using inspection level S-2 and an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Unitization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay as monobasic sodium phosphate. Weigh to the nearest milligram (mg) approximately 5 g of the specimen and dissolve in the minimum amount of water necessary. Using a pH meter accurate to 0.05 pH units, titrate with 1N sodium hydroxide solution to a pH of 9.00. Then titrate the solution with 1N hydrochloric acid to a pH of 4.30. Calculate the percent by weight monobasic sodium phosphate (NaH_2PO_4) as follows:

If AB is greater than CD,

$$\text{Percent monobasic sodium phosphate} = \frac{12(2CD - AB)}{W}$$

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If CD is greater than or equal to AB,

$$\text{Percent monobasic sodium phosphate} = \frac{12AB}{W}$$

where: A = Milliliters of sodium hydroxide solution used in the first titration,
B = Normality of the sodium hydroxide solution,
C = Milliliters of hydrochloric acid used in the second titration,
D = Normality of the hydrochloric acid, and
W = Weight of specimen in grams.

4.2.4.2 Insoluble matter. Weigh to the nearest milligram approximately 20 g of the specimen into a 600-milliliter (mL) beaker. Add 400 mL of water, boil the solution for 10 minutes, and filter through a tared filter crucible. Wash thoroughly with hot water and then dry the crucible and residue at 105° to 110°C for 3 hours. Cool to room temperature in a desiccator and weigh to the nearest 0.1 mg. Calculate the percent by weight insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100A}{W}$$

where: A = Weight of residue, in grams, and
W = Weight of specimen in grams.

4.2.4.3 pH. Determine the pH of a 1.0 percent by weight solution of the specimen in water at 20° ± 5°C using a pH meter accurate to 0.05 pH units.

5. PACKAGING

5.1 Unit packing. Sodium phosphate shall be unit packed level A, B or industrial as specified (see 6.2).

5.1.1 Level A. A quantity of 100 (+1 or -0) pounds (lb) of sodium phosphate shall be unit packed in a removable head steel drum finished with a polyethylene bag liner. The drum shall conform to Rule 40 of the Uniform Freight Classification, shall be of minimum volume to contain the 100 lb of sodium phosphate, and shall be clean, phosphatized, painted, and lined in accordance with PPP-D-729. The bag liner shall conform to type I, class B, style 1 or 2 of MIL-B-117 and shall be sized to fit the drum without stretching. The bag shall be closed by any suitable means. The drum shall be closed by bolted locking ring, using lock washers or self locking nuts on the bolt.

5.1.2 Level B. A quantity of 100 (+1 or -0) lb of sodium phosphate shall be packed in a multiwall paper sack conforming to type optional, style B, sack construction number 15 or 15X, moisture barrier grade MB2 of UU-S-48. The sack shall be closed as specified in UU-S-48.

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5.1.3 Industrial. Uniform quantities of sodium phosphate, as specified in the contract or purchase order, shall be unit packed in accordance with MIL-STD-1188 (see 6.2).

5.2 Packing. Sodium phosphate, unit packed as specified in 5.1, shall require no further protection for shipment other than unitization.

5.3 Unitization. Uniform quantities of level A and B packs of sodium phosphate shall be palletized in accordance with the applicable requirements of MIL-STD-147. Fiberboard cap storage aid 4 shall be used over the load of sacks. Industrial packs shall be unitized in a manner to protect the load from supply source to first destination, to assure carrier acceptance and to comply with Uniform Freight Classification Rules, National Motor Freight Classification Rules and the rules of any other applicable mode of transportation. Unitized loads shall be capable of safe and stable stacking.

5.4 Marking. Level A and B containers and pallet loads shall be marked in accordance with MIL-STD-129. Industrial containers shall be marked in accordance with MIL-STD-1188. Each container shall be marked to show the date of manufacture and lot number of the sodium phosphate.

6. NOTES

6.1 Intended use. Anhydrous monobasic sodium phosphate is intended for use in boiler-water treatment and in electroplating processes.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification, the level of unit packing required (see 5.1), and the quantity per unit container required (see 5.1.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-B378

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

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